$\boxed{T12}$ "4": $8/24$ "3": $4/24$ "2" "1"

$p_4 = 1/3$ $p_3 = 1/6$ $p_2 = p_1 = \frac{1}{4}$

$1 = \frac{1}{3} + \frac{1}{6} + 2p_1 = \frac{1}{2} = \frac{6+12}{24} + p_1 \Rightarrow p_1 = p_2 = \frac{6}{24}$

2 подряд орла $\vec{x}_2$ $n=2$

$H_0: \xi \sim \frac{1}{4}\{1,2\} + \frac{1}{6}\{3\} + \frac{1}{3}\{4\}$

$H_1: \xi \sim \frac{1}{4}\{1,2,3,4\}$

$\alpha = 0{,}2$

H_0	1	2	3	4
1	$1/16$	$1/16$	$\frac{1}{24}$	$1/12$
2	$1/16$	$\frac{1}{16}$	$\frac{1}{24}$	$\frac{1}{12}$
3	$\frac{1}{24}$	$\frac{1}{24}$	$\frac{1}{36}$	$\frac{1}{18}$
4	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{18}$	$\frac{1}{9}$

H_1	1	2	3	4
1	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$
2	$\frac{1}{16}$			
3	$\frac{1}{16}$			
4	$\frac{1}{16}$			$\frac{1}{16}$

$G: l \geq C$ $P(l \geq C \mid H_0) \leq \alpha = 0{,}2$

$l = \frac{L_1}{L_0}$

l_i	1	2	3	4
1	1	1	$3/2$	$3/4$
2	1	1	$3/2$	$3/4$
3	$3/2$	$3/2$	$9/4$	$9/8$
4	$3/4$	$3/4$	$9/8$	$9/16$

Самые большие l_i:

1) $9/4$ $(3,3) \to \frac{1}{36}$

2) $3/2$ $(1,3)(2,3)(3,1)(3,2) \to \frac{1}{6}$

3) $9/8$ $(3,4)(4,3) \to \frac{1}{9}$

$\frac{1}{36} < \alpha = 0,2$

$\frac{1}{36} + \frac{1}{6} = \frac{7}{36} \approx 0,194 < 0,2$

$\frac{1}{36} + \frac{1}{6} + \frac{1}{9} \approx 0,306 > 0,2$

$\to \frac{9}{8} < C \leq \frac{3}{2} \to C = \frac{3}{2}$

$G: \ l \geq \frac{3}{2}$

$W = 5 \cdot \frac{1}{16} = \frac{5}{16}$ $\alpha_1 = \frac{7}{36}$ $\alpha_2 = 1 - W = \frac{11}{16}$